

EXPERIMENT – 1

Design and configure the hub network using cisco packet tracer

AIM : To design and configure a simple network using **one Hub and five PCs** in **Cisco Packet Tracer**, and to assign Class A IP addresses to each PC in order to verify basic network connectivity.

PROCEDURE:

1. Open Cisco Packet Tracer
 - Launch the Cisco Packet Tracer application.
2. Add Network Devices
 - From the Network Devices section, select Hubs.
 - Drag and drop one Hub into the workspace.
3. Add End Devices
 - From the End Devices section, drag and drop five PCs (PC0 to PC4) into the workspace.
4. Connect the Devices
 - Select Copper Straight-Through cable.
 - Connect each PC to the Hub using the FastEthernet ports.
 - § PC0 → Hub
 - § PC1 → Hub
 - § PC2 → Hub
 - § PC3 → Hub
 - § PC4 → Hub

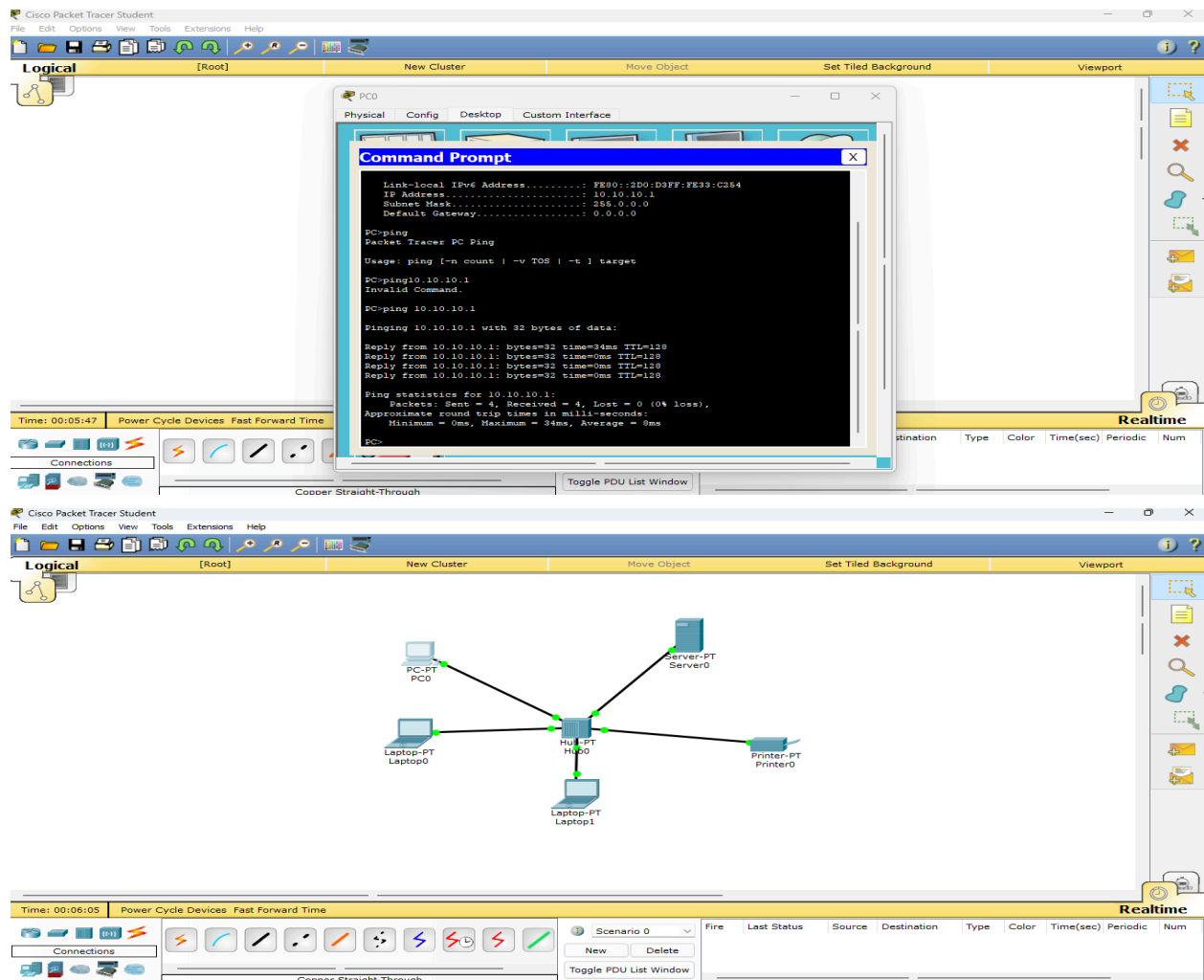
5. Configure IP Addresses

- Click on **PC0** → Desktop → IP Configuration
 - § IP Address: 10.0.0.1
 - § Subnet Mask: 255.0.0.0
- Click on **PC1** → Desktop → IP Configuration
 - § IP Address: 10.0.0.2
 - § Subnet Mask: 255.0.0.0
- Click on **PC2** → Desktop → IP Configuration
 - § IP Address: 10.0.0.3
 - § Subnet Mask: 255.0.0.0
- Click on **PC3** → Desktop → IP Configuration
 - § IP Address: 10.0.0.4
 - § Subnet Mask: 255.0.0.0
- Click on **PC4** → Desktop → IP Configuration
 - § IP Address: 10.0.0.5
 - § Subnet Mask: 255.0.0.0

6. Verify Connectivity

- Open Command Prompt on any PC.
- Use the `ping` command to ping other PCs (e.g., `ping 10.0.0.2`).
- Observe successful replies.

OUTPUT:



RESULT:

A network consisting of **one Hub and five PCs** was successfully designed and configured using Class A IP addresses in Cisco Packet Tracer.

